

MATH 353 Complex Calculus

Credit: (4-0) 4

Catalog description: Algebra of complex numbers. Polar representation. Analyticity. Cauchy-Riemann equations. Power series. Elementary functions. Mapping by elementary functions. Linear fractional transformations. Line integral. Cauchy-Theorem. Cauchy integral formula. Taylor's Series. Laurent series. Residues. Residue theorem. Improper integrals.

Course Objectives: By the end of this course, a student will:

- Understand in detail the complex number system and the complex plane,
- Use functions of a complex variable and explore their properties
- Understand the concepts of the derivative of a complex function and analyticity of a function,
- Make calculations with elementary functions of a complex variable
- Evaluate line integrals and prove results using the Cauchy-Goursat theorem and the Cauchy integral formulas,
- Master the concepts of complex sequences and infinite series, Laurent series, residues and the residue theorem and apply them to various problems
- Understand the basics of conformal mappings and Möbius transformations

Course Website: <https://metuclass.metu.edu.tr/>

Textbook: “Complex Variables and Applications”, Brown, J. W., Churchill, R. V., 9th ed.

Reference Material: “Complex Analysis”, Gamelin, T. W.
“Complex Variables”, Fisher, S. D.

Exams and Grading:

Midterm 1 : 30 % 20 November 2026, Friday, 18:00

Midterm 2 : 30 % 25 December 2026, Friday, 18:00

Final : 40 % (date to be announced)

Suggested Problems: A list of suggested problems will be announced on METUClass. Students are encouraged to attempt to solve all of these problems in a timely manner and discuss their solutions with the instructor.

Attendance Policy: Effective Week 3, students are required to maintain at least 60% attendance. Students who don't satisfy this requirement will not be able to take the final exam and get an NA grade from the course.

Any instance of attendance misconduct will lead to an 'NA' grade for both the party committing the act and the party soliciting it. Report submissions do not count toward the 60% attendance requirement.

Exceptions: Students repeating the course who have previously achieved a grade of **DD or above** are exempt from mandatory attendance. Other special situations may be evaluated individually on a case-by-case basis with the consent of the instructor.

NA Policy: A student who misses all exams OR who doesn't satisfy the attendance requirement will receive a grade of NA for the course.

Make-up Policy: In order to be eligible to enter a make-up examination for a missed examination, a student should have a documented or verifiable, and officially acceptable excuse. A student cannot get make-up examinations for two missed exams. The make-up examination for all exams will be after the final exam, and will include all topics.

Lectures:

Section, Instructor	Lecture Time and Place	Instructor e-mail, Office (Math building), office phone
S1. İbrahim Ünal	Mon 8:40-10:30 (M-13) Wed 10:40-12:30 (M-13)	iunal@metu.edu.tr M-241

Office Hours: Monday 10:40-12:30

Important Dates:

- **September 28:** Classes begin
- **October 5-9:** Add-drop and advisor approvals
- **October 29:** Republic Day (Thursday)
- **November 10:** Commemoration of Atatürk (Tuesday)
- **November 30-December 6:** Course withdrawal applications
- **January 1:** New Year's Day (Friday)
- **January 4-16:** Final exams period
- **January 25:** The announcement of letter grades

Course Schedule

The table below is a rough guideline for the content of course lectures. Instructors may reorder their lectures as necessary/desired. Chapter and subsection titles are from the textbook, *Complex Variables and Applications*, Brown, J. W., Churchill, R. V., 9th ed. Each numbered box below indicates a 2 lecture-hour day, therefore there will be a total of 54 lecture-hours.

Week 1: Sep.28-Oct.2	1	Chapter 1. Complex Numbers Sums and Products, Basic Algebraic Properties, Further Properties, Vectors and Moduli, Triangle Inequality, Complex Conjugates.
	2	Exponential form, Products and Powers in Exponential Form, Arguments of Products and Quotients, Roots of Complex Numbers, Examples, Regions in the Complex Plane.
Week 2: Oct.5-9	3	Chapter 2. Analytic Functions Functions and Mappings, The Mapping $w=z^2$.
	4	Limits, Theorems on Limits, Limits Involving the Point at Infinity, Continuity.
Week 3: Oct.12-16	5	Derivatives, Rules for Differentiation, Cauchy-Riemann Equations, Examples, Sufficient Conditions for Differentiability.
	6	Polar Coordinates, Analytic Functions, Further Examples, Harmonic Functions, Uniquely Determined Analytic Functions, Reflection Principle.
Week 4: Oct.19-23	7	Chapter 3. Elementary Functions The Exponential Function, The Logarithmic Function, Examples, Branches and Derivatives of Logarithms, Some Identities Involving Logarithms.
	8	The Power Function, Examples, The Trigonometric Functions $\sin(z)$ and $\cos(z)$, Zeros and Singularities of Trigonometric Functions, Hyperbolic Functions, Inverse Trigonometric and Hyperbolic Functions.
Week 5: Oct.26-30	9	Chapter 8. Mapping by Elementary Functions Linear Transformations, The Transformation $w=1/z$, Mappings by $1/z$, Linear Fractional Transformations.

Week 5: Oct.26-30	10	An Implicit Form, Mappings of the Upper Half Plane, Examples, Mappings by the Exponential Function, The Transformation $w=\sin z$, Some Related Mappings, Mappings by z^2 and Branches of $z^{1/2}$
Week 6: Nov.2-6	11	Square Roots of Polynomials, Riemann Surfaces, Surfaces for Related Functions.
	12	Chapter 4. Integrals Derivatives of Functions $w(t)$, Definite Integrals of Functions $w(t)$, Contours, Contour Integrals, Some Examples.
Week 7: Nov.09-13	13	Examples Involving Branch Cuts, Upper Bounds for Moduli of Contour Integrals, Antiderivatives, Proof of the Theorem.
	14	Cauchy-Goursat Theorem, Proof of the Theorem, Simply Connected Domains, Multiply Connected Domains, Cauchy Integral Formula.
Week 8: Nov.16-20	15	An Extension of the Cauchy Integral Formula, Verification of the Extension, Some Consequences of the Extension.
	16	Liouville's Theorem and the Fundamental Theorem of Algebra, Maximum Modulus Principle.
Week 9: Nov.23-27	17	Chapter 5. Series Convergence of Sequences, Convergence of Series, Taylor Series, Proof of Taylor's Theorem.
	18	Examples, Negative Powers of $(z-z_0)$, Laurent Series, Proof of Laurent's Theorem, Examples.
Week 10: Nov.30-Dec.4	19	Absolute and Uniform Convergence of Power Series, Continuity of Sums of Power Series.
	20	Integration and Differentiation of Power Series.
Week 11: Dec.7-11	21	Uniqueness of Series Representations, Multiplication and Division of Power Series.
	22	Chapter 6. Residues and Poles Isolated Singular Points, Residues, Cauchy's Residue Theorem, Residue at Infinity.
Week 12: Dec.14-18	23	The Three Types of Isolated Singular Points, Examples, Residues at Poles, Examples.
	24	Zeros of Analytic Functions, Zeros and Poles, Behavior of Functions Near Isolated Singular Points.
Week 13: Dec.21-25	25	Chapter 7. Applications of Residues Evaluation of Improper Integrals, Example, Improper Integrals from Fourier Analysis.
	26	Jordan's Lemma, An Indented Path, An Indentation Around a Branch Point, Integration Along a Branch Cut.
Week 14: Dec.28-30	27	Definite Integrals Involving Sines and Cosines, Argument Principle, Rouché's Theorem.